# Executive Summary: Credit Risk ML Model

## 1. The Business Challenge

Traditional models optimise for accuracy and apply a 0.5 decision threshold, assuming all errors are equally costly. In credit risk, that assumption is wrong by an order of magnitude. Missing a default costs GBP 25,000 (principal loss + collections), while wrongly rejecting a good customer costs GBP 2,500 (lost interest). This 10:1 cost ratio requires a specialised, cost-sensitive approach.

## 2. Business Result & Financial Impact

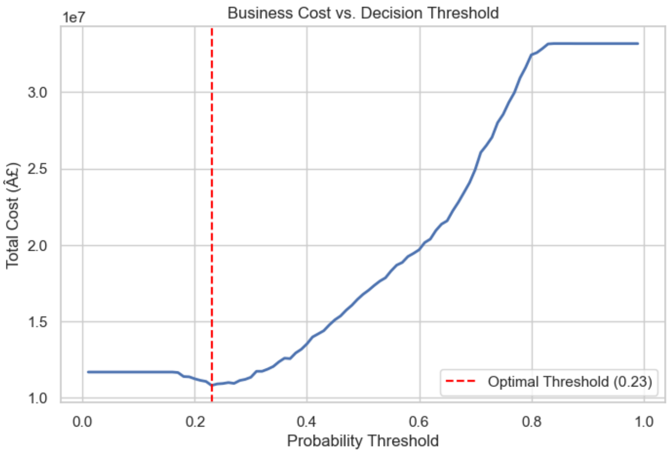
By replacing the naive 0.5 probability threshold with a mathematically derived optimal threshold of 0.23, the XGBoost model transformed abstract ML accuracy into a quantifiable financial outcome:

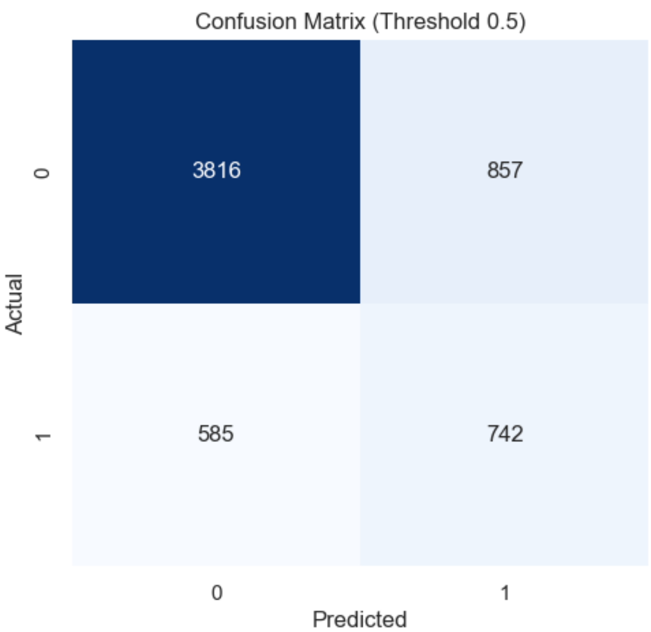
**Impact on the 6,000-customer test portfolio:**

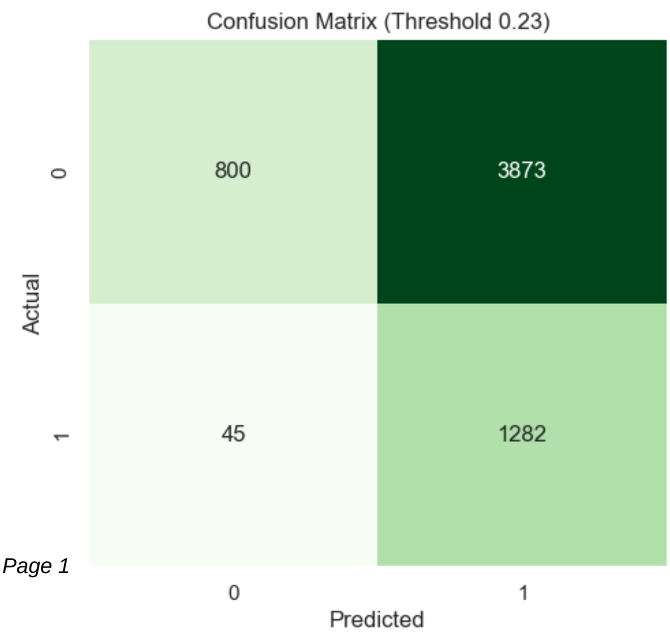
- Defa

The GBP 5.96M figure accounts for every false positive incurred at the lower threshold - the saving is real net of intervention cost.

## 3. Performance & Threshold Analysis



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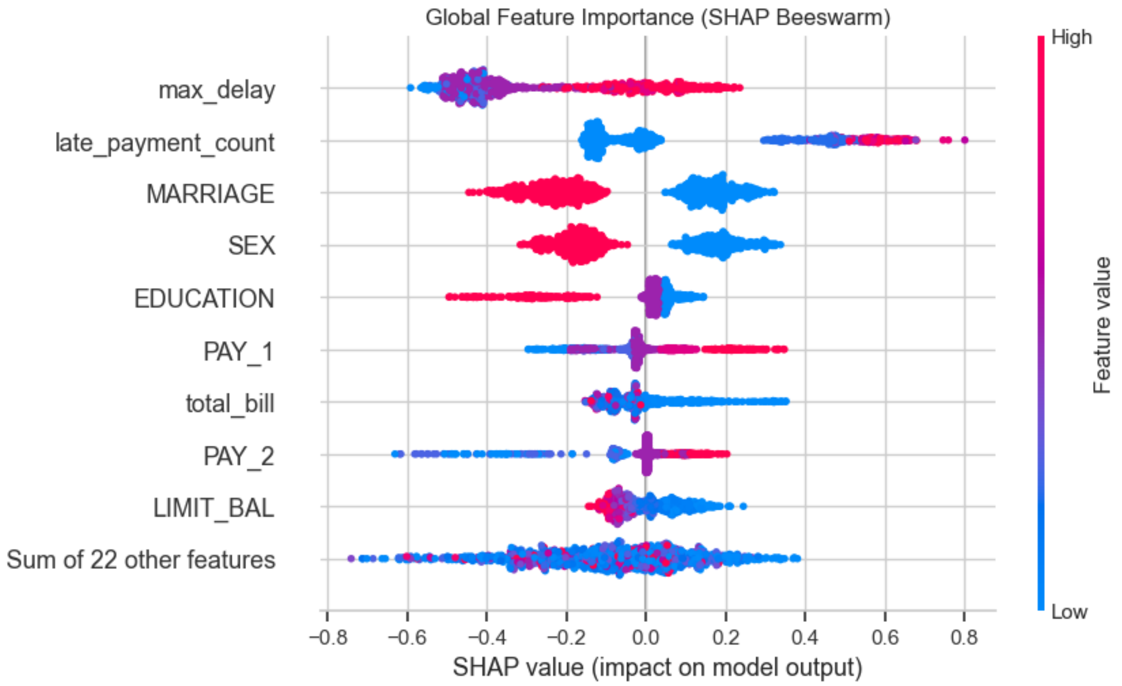
## 4. Explainability and Risk Factors

The model utilizes SHAP (SHapley Additive exPlanations) to provide full transparency. The beeswarm plot below shows the most critical risk drivers across the entire portfolio. Payment delays (PAY_1), low limit balances, and high utilization rates are the strongest indicators of default.




## 5. Conclusion & Recommendations

The cost-sensitive model successfully bridges the gap between technical metrics and business ROI. We recommend deploying this model in a shadow-testing environment to validate the projected GBP 5.96M savings per 6,000 applicants, while actively monitoring the top SHAP features for macroeconomic drift.